WORKED SOLUTION

STEP I 2003, Question 5

Pure • Binomial Series

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s1-q5

The key idea

In any binomial expansion, write down the general term and track the power of x it carries; the constant term is the one where that power is zero. In part (ii) the expansion is not a finite one, so first pull out the dominant factor: for large $|x|$ that is x^6, leaving a series in $3/x$; for small $|x|$ it is $3x^5$, leaving a series in $x/3$. Which powers of x can then appear settles everything.

Solution

Part (i)

The general term of $(2x + x^{-2})^6$ is

$$\binom{6}{r}(2x)^{6-r}(x^{-2})^r = \binom{6}{r}2^{6-r}x^{6-3r}, \qquad r = 0, 1, \ldots, 6.$$

This is independent of x exactly when $6 - 3r = 0$, that is $r = 2$, and then the term is

$$\binom{6}{2}2^4 = 15 \times 16 = 240,$$

as required.

In the same way, the general term of $(ax^3 + bx^{-2})^{5n}$ is

$$\binom{5n}{r}(ax^3)^{5n-r}(bx^{-2})^r = \binom{5n}{r}a^{5n-r}b^r\, x^{15n-5r}.$$

The power of x is zero when $15n = 5r$, so $r = 3n$ (which lies in $0 \leq r \leq 5n$). The term independent of x is therefore

$$\binom{5n}{3n}a^{2n}b^{3n} = \frac{(5n)!}{(2n)!\,(3n)!}\,a^{2n}b^{3n}.$$

Part (ii)

Large $|x|$. Taking out the factor x^6,

$$f(x) = (x^6(1+3/x))^{1/2} = x^3\Big(1+\frac{3}{x}\Big)^{1/2},$$

taking $(x^6)^{1/2} = x^3$ as the question intends (strictly this is $|x|^3$; see the note below). For $|x| > 3$ we have $|3/x| < 1$, so the binomial series for $(1+u)^{1/2}$ with $u = 3/x$ converges:

$$(1+u)^{1/2} = 1 + \tfrac{1}{2}u - \tfrac{1}{8}u^2 + \tfrac{1}{16}u^3 - \tfrac{5}{128}u^4 + \cdots,$$

where the u^3 coefficient is

$$\binom{1/2}{3} = \frac{\frac{1}{2}\cdot(-\frac{1}{2})\cdot(-\frac{3}{2})}{3!} = \frac{3/8}{6} = \frac{1}{16}.$$

Multiplying the series by x^3, the term $\binom{1/2}{k}3^k x^{-k}$ becomes $\binom{1/2}{k}3^k x^{3-k}$, so only $k = 3$ gives a constant:

$$f(x) = x^3 + \tfrac{3}{2}x^2 - \tfrac{9}{8}x + \frac{1}{16}\cdot 27 - \frac{405}{128x} + \cdots,$$

and the term independent of x is $\frac{27}{16}$, as required.

Ask yourself What happens when $x < -3$?

The square root means the non-negative root, and $(x^6)^{1/2} = |x|^3$, which equals $-x^3$ when $x < 0$. So for $x < -3$ every term of the expansion changes sign, and the constant term is $-\frac{27}{16}$. The stated result $\frac{27}{16}$ is exactly right for $x > 3$; for $x < -3$ it holds if we read $(x^6)^{1/2}$ as x^3, which is evidently what the question intends.

Small $|x|$. Now the dominant term inside the bracket is $3x^5$, so we take that out instead:

$$x^6 + 3x^5 = 3x^5\Big(1+\frac{x}{3}\Big).$$

For $f(x)$ to be real we need $x^5(x+3) \geq 0$; with $|x| < 3$ the factor $x+3$ is positive, so we need $x \geq 0$. (For $-3 < x < 0$, f isn't defined as a real function at all.) For $0 < x < 3$,

$$f(x) = \sqrt{3}\,x^{5/2}\Big(1+\frac{x}{3}\Big)^{1/2} = \sqrt{3}\,x^{5/2}\sum_{k\geq 0}\binom{1/2}{k}\frac{x^k}{3^k},$$

and the series converges since $|x/3| < 1$. Every term is a constant multiple of

$$x^{5/2+k}, \qquad k = 0, 1, 2, \ldots,$$

and $\frac{5}{2} + k$ is never an integer, so in particular never 0. Hence the expansion of $f(x)$ in powers of x for $|x| < 3$ has no term independent of x. (At $x = 0$ itself, $f(0) = 0$.)

Checks

- For large positive x, $f(x) - \left(x^3 + \frac{3}{2}x^2 - \frac{9}{8}x\right)$ should approach $\frac{27}{16} = 1.6875$. At $x = 1000$ a calculator gives 1.68434, and the next term of the series, $-\frac{405}{128x} \approx -0.00316$, accounts for the gap.
- Near 0, $f(x) \approx \sqrt{3}\,x^{5/2}$: at $x = 0.01$, $f(0.01) = \sqrt{10^{-12} + 3 \times 10^{-10}} \approx 1.734 \times 10^{-5}$, close to $\sqrt{3} \times 10^{-5}$, and certainly not tending to a non-zero constant.
- For the general formula, $n = 1$, $a = b = 1$ gives $\binom{5}{3} = 10$, the constant term of $(x^3 + x^{-2})^5$; expanding directly confirms it.

Common slips

- Writing the general term with the powers the wrong way round, e.g. $(2x)^r(x^{-2})^{6-r}$, and then mismatching r with the binomial coefficient. Either labelling works if you are consistent.
- Getting the sign of $\binom{1/2}{3}$ wrong: two negative factors $(-\frac{1}{2})(-\frac{3}{2})$ make it positive.
- Expanding $(x^6 + 3x^5)^{1/2}$ for small x by taking out x^6: that gives a series in $3/x$, which diverges when $|x| < 3$.

Beyond STEP

The two expansions of f are valid on different regions, and which factor you pull out decides which region you get. This is the idea of expanding a function about different points, and it leads to Laurent series and to Puiseux series (expansions in fractional powers, like $x^{5/2+k}$ here) in complex analysis. The branch of the square root that caused the sign issue for $x < -3$ is also a central theme there. Needham's *Visual Complex Analysis* explains both pictorially.

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